

Self-Distillation of XLM-RoBERTa for Multilingual Sentiment Analysis on Imbalanced Data: A Case Study on Malagasy Mobile Application Reviews

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Abstract—This study proposes a self-distillation framework based on XLM-RoBERTa-base for binary sentiment analysis of multilingual application reviews in French, English, and Malagasy. To address class imbalance and low-resource challenges, the framework integrates high-temperature soft-label transfer, feature-level representation alignment, and focal-loss-based class weighting. The self-distilled student model outperformed its XLM-RoBERTa teacher baseline, achieving 91.39% accuracy (+0.99%) and 90.63% macro-F1 (+1.10%), with consistent gains in negative-class detection. Cross-lingual evaluation revealed asymmetric transfer effects, with improved performance for French by 1.75%, remained stable for Malagasy, and decreased for English by 3.45%, likely reflecting domain differences in English training sample distribution. These results demonstrate that self-distillation effectively regularizes transformer models under imbalanced multilingual conditions, with cross-lingual generalization being most effective when sufficient target-language representation is available during training.

Keywords—Knowledge distillation, low-resource languages, multilingual, sentiment analysis, XLM- RoBERTa.

I. INTRODUCTION

Reviews of mobile applications have become one of the most important ways for customers to give feedback to developers [1]. However, reviews in Madagascar are rarely written in a single language. Many users usually mix French, Malagasy, and English [2], reflecting the country's multilingual reality. Analyzing the sentiment of such mixed-language text requires powerful multilingual models like XLM-RoBERTa, which has shown strong performance across multiple languages [4], [5]. However, despite their strong predictive capabilities, the performance of these massively multilingual models severely degrade when applied to highly imbalanced, low-resource datasets, often resulting in a strong bias toward the majority class [6], [7].

There are two main challenges in this situation. First, there is an imbalance in the classes, in which 66.5% of reviews are positive, and only 33.5% are negative, causing models to misclassify minority-class samples and reducing

generalization [8]. Second, Malagasy is a low-resource language, facing the same data scarcity challenges documented for other underrepresented languages, which makes it harder for models to learn it effectively [9], [10].

Knowledge Distillation (KD) is primarily studied for model compression using different architectures for the teacher and the student. Its application as self-distillation with identical architectures, particularly under severe class imbalance and low-resource multilingual conditions involving languages such as Malagasy, remains insufficiently explored. This study evaluates whether an XLM-RoBERTa self-distillation framework can improve minority-class and overall performance in imbalanced multilingual sentiment classification and examines cross-lingual transfer effects across French, English, and Malagasy.

This paper makes four primary contributions. First, it presents a novel application of self-distillation using identical XLM-RoBERTa architectures for both the teacher and student models, specifically designed for sentiment analysis in a low-resource multilingual context. Second, it reformulates knowledge distillation as an algorithmic regularizer by integrating high-temperature soft-label transfer, feature alignment, and focal loss to mitigate minority-class bias. Third, it provides empirical evidence demonstrating that the distilled student model surpasses the teacher baseline in overall performance while also improving the detection of the minority negative class. Finally, the paper presents a cross-lingual evaluation that reveals asymmetric transfer effects across high-resource languages and the morphologically distinct low-resource language Malagasy.

II. RELATED WORK

A. Multilingual Sentiment Analysis and XLM- RoBERTa-base

Sentiment analysis has increasingly shifted from high-resource monolingual corpora [11] to complex, code-mixed environments [12]. Transformer-based models such as BERT

have demonstrated strong classification performance across various text analysis tasks, including distinguishing risk from crisis communication in social media contexts [3]. Building on this foundation, Massively Multilingual Transformers (MMTs) like XLM-RoBERTa-base have demonstrated strong zero-shot transfer capabilities in resource-poor Indian languages [4] and structured Korean text [13]. However, applying them to severely underrepresented and non-standardized languages like Malagasy remains challenging, primarily due to a lack of foundational datasets [9], [10]. Furthermore, while XLM-RoBERTa-base yields strong predictive metrics in multi-language tasks [5] its effectiveness in localized, code-mixed contexts is often hindered by target-language scarcity.

B. Handling Class Imbalance in Sentiment Classification

Compounding resource scarcity is the severe class imbalance found in real-world sentiment datasets. Positive reviews overwhelmingly dominate, forcing models to over-index on majority patterns [6], significantly degrading minority-class recall and leading to frequent misclassification of critical negative reviews [7]. Traditional solutions, such as SMOTE or random oversampling [14] are inefficient for high-dimensional transformer embeddings, necessitating the algorithmic self-distillation approach proposed in this study.

C. Self-Distillation

Knowledge distillation mainly transfers knowledge from a large teacher to a smaller student model [15], [16]. Self-distillation instead uses identical architectures for both teacher and student to improve learning and generalization [17] amplifying regularization without changing model size [18]. It was selected here because it improves minority-class sensitivity without requiring a larger teacher model or data augmentation, making it the most practical choice for imbalanced, low-resource multilingual settings. Despite this potential, self-distillation has rarely been applied to imbalanced, low-resource multilingual settings, a gap this paper addresses.

III. METHODOLOGY

This section describes the pipeline for multilingual binary sentiment analysis of Mvola reviews from Google Play across French, English, and Malagasy, using XLM-RoBERTa Base. The proposed framework is illustrated in Fig. 1.

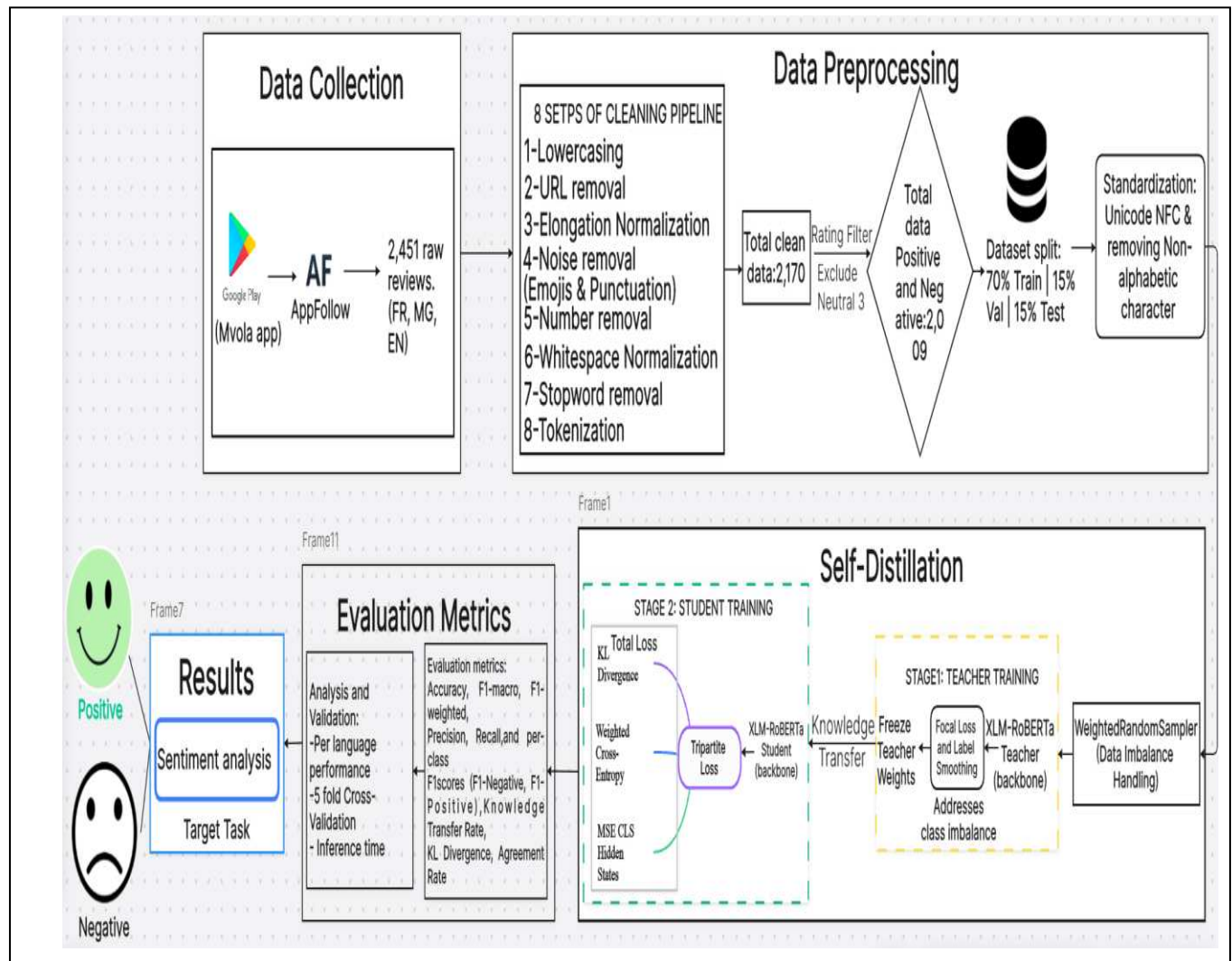


Fig. 1. Framework overview

A. Data Collection

This study targeted the review section of the Mvola application, a widely used mobile payment platform in Madagascar. Data were collected using the AppFollow tool, yielding 2,451 raw reviews in French (FR), English (EN), and Malagasy (MG), often in code-mixed formats.

B. Data Preprocessing

The 2,451 raw reviews were initially screened for missing and empty entries, which were subsequently discarded. The remaining text was processed through an eight-step preprocessing pipeline. Specifically, the pipeline performed lowercasing, URL removal, elongation normalization, noise removal of emojis and punctuation, number removal, whitespace normalization, multilingual stopword removal covering French, English, and Malagasy, and finally tokenization using the XLM-RoBERTa subword tokenizer [5] with Byte-Pair Encoding (BPE), applying a maximum sequence length of 128 tokens. This pipeline yielded 2,170 clean samples for subsequent analysis.

Reviews with a neutral rating of 3 were excluded (161 samples, 7.4%) to avoid label ambiguity. The remaining 2,009 samples were mapped to binary labels, where ratings 1–2 were designated as Negative and ratings 4–5 as Positive, resulting in 673 negative samples (33.5%) and 1,336 positive samples (66.5%). The dataset was split into 70% training, 15% validation, and 15% test sets using stratified partitioning. A WeightedRandomSampler was applied to increase sampling probability for the minority class. Following preprocessing, cleaned tokens were reassembled into normalized sequences by stripping residual whitespace, removing non-alphabetic characters, and applying Unicode NFC normalization.

C. Proposed Self-Distillation Framework

The framework fine-tunes XLM-RoBERTa base [5] on our trilingual dataset, reframing knowledge distillation as algorithmic regularization [18] rather than model compression. Both the Teacher and Student models share identical XLMRobertaClassifier architectures, comprising a classification head of layer normalization, dropout layer with rate of 0.3, a fully connected layer from 768 to 256, GELU activation, dropout with rate of 0.15, and a final fully connected layer with 2 output classes. class weights, computed as the total number of samples divided by the product of the number of classes and samples per class, are incorporated into both loss functions.

The Teacher model is trained using focal loss with a focusing parameter (γ) of 2.0 and label smoothing with a smoothing factor (ϵ) of 0.1. The student model is trained using a tripartite loss, formally defined in Eq. (1):

$$L = \alpha \cdot LKD + \beta \cdot LCE + \gamma \cdot LFeat \quad (1)$$

The total loss (L) is computed as the weighted sum of three components. Specifically, LKD refers to the knowledge distillation loss, computed as the Kullback–Leibler (KL) divergence between the teacher and student output distributions scaled by a temperature (T) of 6.0, weighted by a coefficient alpha (α) of 0.60, LCE corresponds to the weighted cross-entropy loss computed against the hard ground-truth labels, controlled by a coefficient beta (β) of

0.25, and LFeat denotes the feature alignment loss, computed as the Mean Squared Error (MSE) between the teacher and student hidden representations at the [CLS] token level, weighted by a coefficient gamma (γ) of 0.15. Teacher weights are frozen before student training begins. Fig. 2 illustrates the two-stage self-distillation framework, where a trained XLM-RoBERTa teacher transfers knowledge to an identical student network for trilingual sentiment analysis.

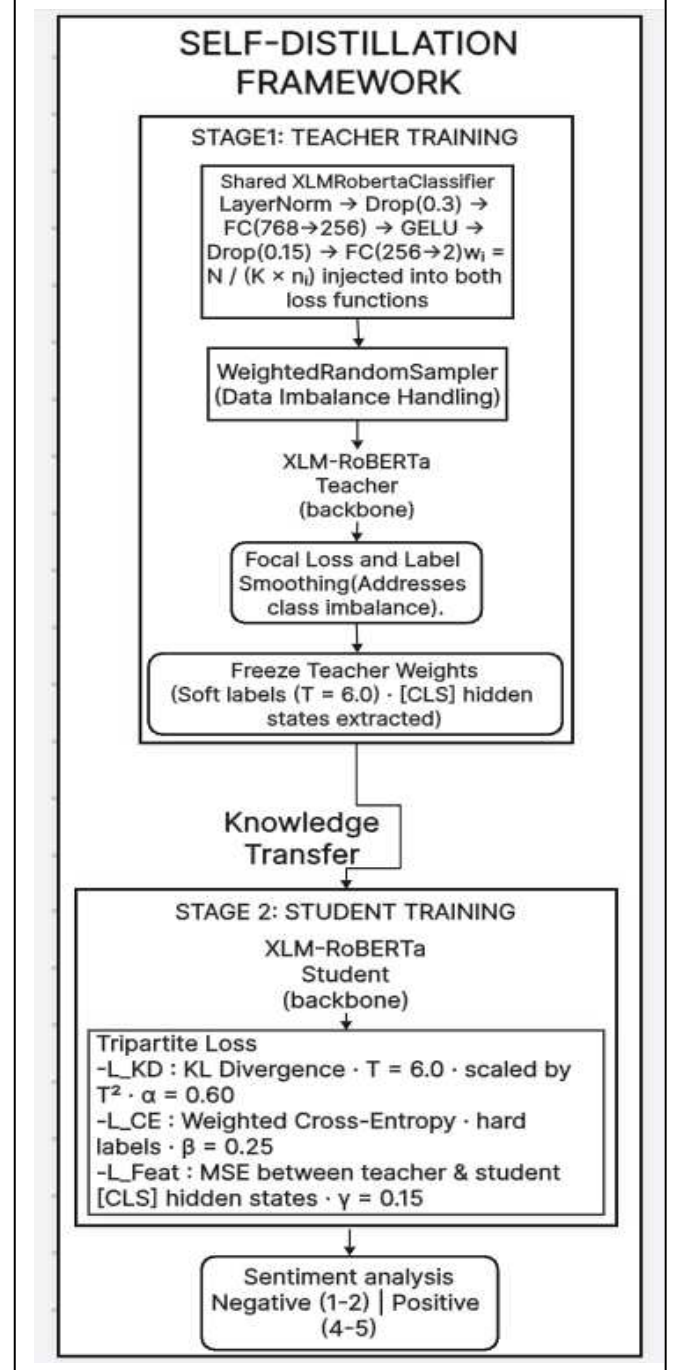


Fig. 2. Self-Distillation framework

D. Training configuration

Both models are optimized using the AdamW optimizer with an epsilon (ϵ) of $1e-8$, a weight decay of 0.01, and a learning rate of $2e-5$. Training is conducted with a batch size of 16, a cosine learning rate schedule, and a 10% linear

warm-up phase. The classification head employs GELU as the activation function, and gradient clipping with a maximum norm of 1.0 is applied to prevent exploding gradients. Both models are trained for a maximum of 100 epochs, with early stopping triggered after 15 consecutive epochs of no improvement, monitored on the validation F1-macro score using a focal loss combined with a knowledge distillation composite loss function.

E. Evaluation Metrics

Model performance is evaluated on the held-out test set (15%) using accuracy, F1-macro, F1-weighted, precision-macro, recall-macro, and per-class F1 scores encompassing F1-Negative and F1-Positive, computed via scikit-learn. Inference time (ms/sample) is recorded for both the teacher and student models. Distillation-specific metrics include the Knowledge Transfer Rate (KTR), defined as the ratio of student accuracy to teacher accuracy, and the Agreement Rate. KL divergence is employed to measure distributional alignment between the teacher and student output distributions. Performance is reported separately for each language, namely French, English, and Malagasy, and generalization is further validated through 5-fold stratified cross-validation.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Results

Table I illustrates representative samples of the standardized dataset (clean_text_standardized) across the three languages and both sentiment classes.

TABLE I. REPRESENTATIVE DATASET SAMPLES

Languages	Clean_text_standardized	Sentiment
English	mvola network fails often works still brag about country wishful thinking best	Negative
English	Great app	Positive
French	regrette avoir compte complications secourir temps voulu	Negative
French	facile pratique	Positive
Malagasy	Tena ratsy ilay app	Negative
Malagasy	Milay app, misaotra telma	Positive

Table II shows the stratified dataset split which consist of 70% training (1,406 samples), 15% validation (302 samples), and 15% testing (301 samples). Each set maintains a consistent class ratio of approximately 66.5% positive and 33.5% negative (1.99:1), meaning for every 673 negative reviews, there are 1,336 positive ones. This imbalance appears across all three sets, confirming the need for class-weighting, WeightedRandomSampler, and focal loss to prevent the model from ignoring the minority negative class.

TABLE II. DATASET SPLIT AND CLASS DISTRIBUTION

Split	Total	Positive (n %)	Negative (n %)	Ratio (Pos:Neg)
Full dataset	2,009	1,336 (66,5%)	673 (33,5%)	1.99: 1
Training data (70%)	1,406	935 (66,5%)	471 (33,5%)	1.99: 1
Validation data (15%)	302	201 (66,6%)	101 (33,4%)	1.99: 1
Test data (15%)	301	200 (66,4%)	101 (33,6%)	1.98: 1

Table III presents the complete performance comparison. The student outperforms the teacher across every metric, achieving 91.39% accuracy (+0.99%), macro-f1 of 90.63% (+1.10%), recall-macro of 92.05% (+1.24%), and precision-macro of 89.70% (+1.04%), confirming the student correctly retrieves more true labels while making fewer incorrect predictions. F1-macro is the most telling metric as it averages performance equally across both classes. the ktr of 1.0110 confirms the student exceeds its teacher, while the agreement rate of 97.68% and near-zero kl divergence of 0.000981 validate successful knowledge transfer.

TABLE III. TEACHER VS STUDENT (TEST SET PERFORMANCE)

Metric	Teacher	Student	Δ (S-T)
Accuracy	90.40%	91.39%	+0.99%
F1-macro	89.53%	90.63%	+1.10%
F1-weighted	90.53%	91.51%	+0.99%
Recall-macro	90.82%	92.05%	+1.24%
Precision-macro	88.66%	89.70%	+1.04%
F1-Negative	86.51%	87.96%	+1.45%
F1-Positive	92.54%	93.30%	+0.75%
Agreement rate	—	97.68%	—
Knowledge transfer rate (KTR)	—	97.68%	—
KL divergence	—	0.000981	—
Inference (ms/sample)	7.11	14.03	+6.92

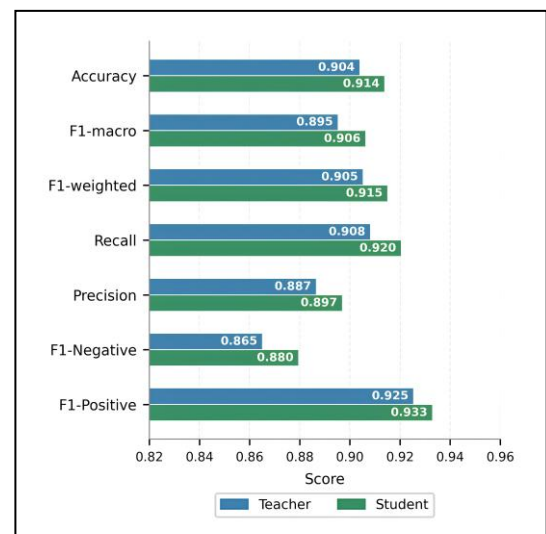


Fig. 2. Performance of the teacher and student on test set across all metrics

The most critical gain is F1-Negative, rising from 86.51% to 87.96% (+1.45%), the hardest class to

detect, given only 33.5% of samples are negative. This confirms the framework achieves its core objective through the tripartite loss, where the teacher develops minority-class sensitivity via focal loss, which the student inherits through soft-label transfer and feature alignment. F1-Positive also improves from 92.54% to 93.30%, confirming balanced gains.

Fig. 4 represents the confusion matrix of the teacher model, whereas Fig. 5 shows the confusion matrix of the student model. The teacher correctly classifies 92.1% of negative reviews and 89.6% of positive reviews. The student improves on both classes, where negative recall rises to 94.1%, meaning fewer negative reviews are missed, and positive recall reaches 90.0%. The false positive rate also drops from 7.9% to 5.9%, meaning fewer positive reviews are incorrectly labeled as negative. These improvements are consistent with the F1-Negative gain in Table III and confirm that the tripartite loss successfully transfers the teacher’s minority-class sensitivity to the student.

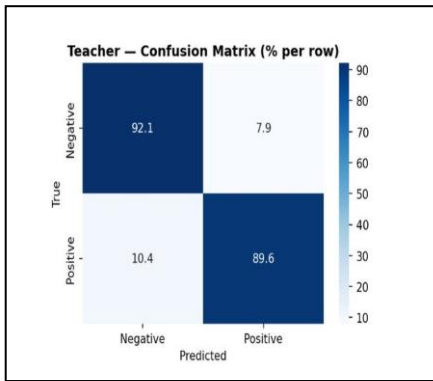


Fig. 3. Teacher confusion matrix

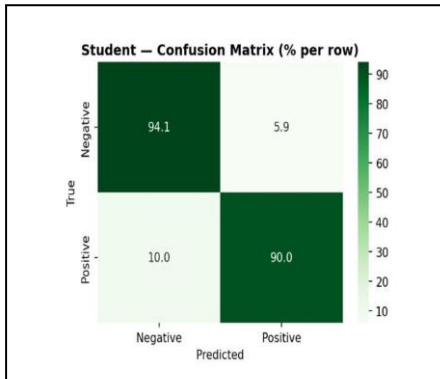


Fig. 4. Student confusion matrix

Table IV reports the per-language accuracy and F1-macro scores for French (n=228), Malagasy (n=45), and English (n=29) on the held-out test set. The results reveal asymmetric transfer effects across the three languages, indicating that the degree of improvement achieved by the student model varies considerably depending on the extent to which each language is represented in the training data.

TABLE IV. PER-LANGUAGE PERFORMANCE ON TEST SET

Language (n)	Teacher Accuracy	Student Accuracy	Teacher F1-macro	Student F1-macro
French (228)	90.35%	92.11%	89.63%	91.52%
Malagasy(45)	86.67%	86.67%	84.45%	85.00%
English (29)	96.55%	93.10%	96.09%	91.94%

French achieves the greatest gain which increased by 1.75% for the accuracy, and by 1.89% for the F1-macro as the dominant training language. Malagasy shows no accuracy improvement, with only a marginal F1-macro gain of 0.55%, attributable to its severe underrepresentation and limited XLM-RoBERTa pre-training coverage. English, conversely, experiences a decline of 3.45% in accuracy and 4.15% in F1-macro, likely due to conflicting signals introduced by French-dominated soft labels, further compounded by its small test set size.

Table V confirms stable generalization across all folds, with F1-Negative consistently below F1-Positive, confirming minority-class detection as the hardest challenge.

TABLE V. 5-FOLD STRATIFIED CROSS-VALIDATION RESULTS

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Accuracy	91.29%	89.80%	90.30%	92.54%	89.03%
F1- macro	90.11%	88.55%	88.85%	91.72%	87.72%
F1- weighted	91.25%	89.79%	90.17%	92.58%	89.05%
Recall- macro	89.74%	88.48%	87.94%	92.18%	87.86%
Precision- macro	90.52%	88.62%	90.03%	91.32%	87.58%
F1- Negative	86.69%	84.76%	84.82%	89.13%	83.70%
F1- Positive	93.53%	92.34%	92.87%	94.32%	91.73%
Average	90.45%	88.90%	89.28%	91.97%	88.09%

Accuracy ranges from 89.03% to 92.54% across the five folds, while F1-macro spans from 87.72% to 91.72%, reflecting consistent performance without overfitting to any specific data split. F1-Negative peaks at 89.13% in Fold 4 and drops to 83.70% in Fold 5, whereas F1-Positive remains comparatively higher, ranging from 91.73% to 94.32%, further reinforcing the inherent difficulty of detecting the minority negative class despite the application of class-weighting, WeightedRandomSampler, and Focal Loss strategies.

B. Discussion

Compared to related work, Kumar and Albuquerque [4] reported 60.93% average accuracy on Hindi using zero-shot transfer learning, and Shin et al. [13] achieved 89.84% accuracy on Korean using XLM-R with architectural modifications. Our student model surpasses both at 91.39% accuracy and 90.63% F1-macro, confirming that self-distillation provides consistent gains over existing approaches.

The results demonstrate that XLM-RoBERTa self-distillation effectively regularizes imbalanced multilingual sentiment analysis, with the agreement rate (97.68%) and near-zero KL divergence confirming successful knowledge transfer, consistent with [18]. However, cross-lingual generalization remains limited, as soft-label transfer amplifies dominant language patterns, explaining the stagnation of Malagasy and the decline of English.

The 6.03% gap between F1-Positive (92.54%) and F1-Negative (86.51%) confirms majority-class bias. Accuracy alone is misleading, making F1-macro the key metric. Three complementary techniques, namely class-weighted loss, WeightedRandomSampler, and focal loss, successfully raised F1-Negative by 1.45%, the greatest improvement observed. Despite these gains, the two-phase training increases inference time to 14.03 ms/sample versus the teacher's 7.11 ms/sample, a practical concern for real mobile deployment where response speed matters.

Beyond Mvola, this framework applies to any multilingual mobile application in Madagascar facing similar challenges, automatically flagging negative reviews across domains such as e-banking or shopping applications. Most importantly, this study supports low-resource languages, ensuring that feedback written in Malagasy or other local languages is finally considered.

V. CONCLUSION AND FUTURE WORK

In this work, a self-distillation framework based on XLM-RoBERTa was proposed to perform binary sentiment classification of code-mixed Google Play Store reviews of the Mvola mobile application in Madagascar, spanning French, English, and Malagasy, with a particular focus on accurately detecting minority negative reviews under severe class imbalance and low-resource multilingual conditions. Using a composite loss combining soft-label transfer, feature alignment, and focal loss, the framework demonstrated that self-distillation effectively regularizes transformer models under imbalanced multilingual conditions, successfully improving minority-class detection while maintaining overall model generalization.

The student outperformed its teacher with a KTR of 1.0110, confirming that self-distillation effectively improves model performance without requiring a larger teacher model or data augmentation. However, cross-lingual evaluation revealed asymmetric transfer effects across the three languages, highlighting the direct influence of language resource availability on generalization performance, particularly for low-resource languages such as Malagasy.

Future work will focus on expanding the Malagasy dataset and extending coverage to other underrepresented languages across Africa and the Indian Ocean region. As the target deployment context involves mobile applications serving speakers of low-resource languages where computational resources are limited and response speed directly affects user experience, applying post-training optimization techniques such as pruning or quantization to the distilled model represents a natural and important next step toward practical real-world deployment.

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